Pythia training dynamics by update range

Pythia-14M Pythia-31M Pythia-70M

Updates 1-100

Updates 100-400

Change in ln(training loss) (5-update trailing mean)

$\hat{\eta}\hat{\kappa}_t$ (proxy)

Raw adjacent cosine ρ_t

Raw gradient $\|g_t\|_2$

Optimizer update

Optimizer update

Loss row: N=5 mean of $\ln L_t - \ln L_{t-1}$; other rows unsmoothed. Seed 1235.

Y limits vary by column; norm row uses log Y. Update 100 appears in both columns; smoothing spans the boundary.

First loss-change mean at update 6; first raw pair undefined. AdamW eta*kappa is a gradient-change proxy.